

UCC302: CIVIL ENGINEERING MATERIALS

L	T	P	Cr
3	0	2	4.0

Course Objective: To expose students to the various building and general construction products and their associated quality, durability and availability.

Cement: Manufacturing, raw ingredients, chemical and mineralogical composition, types of cement, and chemistry of cement hydration

Aggregates: Classification and mechanical properties of aggregate, gradation, effect aggregate type on concrete properties, light weight aggregates

Cement mortar and concrete: Classification, properties and applications, water cement ratio, fresh, mechanical and durability properties of concrete. Production of concrete and mix design.

Chemical and Mineral admixtures: Supplementary Cementitious materials (Fly ash, GGBS, Metakoline, Silica Fume), **Admixtures:** accelerating admixture, water reducing admixture, retarding admixture, air-entraining admixture.

Characterization of cementitious composites (cement and concrete): Characterization techniques with their meaning and uses, microstructure investigation methods including X-ray diffraction, scanning electron microscopy and Thermo-Gravimetric Analysis

Bricks and blocks: Classification and characteristics of bricks, AAC blocks, hollow concrete blocks.

Steel: Market forms of steel, classification and mechanical properties of steel.

Composites materials and nanomaterials in Civil engineering: Introduction to composite laminates – FRP and their use in civil engineering applications. Different types of Nanomaterials in Civil Engineering Applications

Laboratory Work

1. Tests on cement (fineness, specific gravity, consistency and strength test).
2. Tests on fine and coarse aggregates (gradation, specific gravity, water absorption).
3. Design of concrete mix for required grade of concrete.
4. Tests on fresh concrete (workability test).
5. Tests on hardened concrete (compressive strength, flexural strength and split tensile strength).
6. Characterization of cementitious composites using SEM and XRD

Experiential Learning (ELC) Activity

Casting of concrete canoe using innovative lightweight concrete.

Course Learning Objectives (CLO)

The students will be able to:

1. Design the mix proportions for required grade of concrete.
2. Perform various quality control tests for the civil engineering materials in the laboratory
3. Characterize cementitious composites (identify hydration products of cement, interfacial zones in concrete).
4. Evaluate the mechanical properties of steel using uniaxial stress-strain curve

Text Books

1. Gambhir M. L., Concrete Technology, Tata McGraw Hill Publishing Co. Ltd., New Delhi, 2013.
2. Rangawala S. C., Engineering Materials, Charotar Publishing House, Anand, 2005.
3. Chaudhary S. C., Patel K. A., Testing & Evaluation of Civil Engineering Materials, All India Council for Technical Education (AICTE), 2023.

Reference Books

1. Shetty M. S., Concrete Technology: Theory and Practice, S. Chand & Company, 2019.
2. Mehta and Monterio, Concrete Microstructure, Properties and Materials, McGraw-Hill Publishers, 2017.

Evaluation Scheme

Sr. No.	Evaluation elements	Weightage (%)
1	MST	30
2	EST	40
3	Sessional: (May include the following) Assignment, Regular Lab assessmentand Quizzes, Project (Including report, presentation etc.)	30